

Title: Healthy Joints

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenure d (T) or Tenure - track(T T))	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant]
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Corresponding Lead:

Planned start date of seed funding: 1-10-2022

Primary Convergence theme: *Human Centered Technology and AI for Health*Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*Tertiary Convergence theme (if applicable): *Improving Health Journeys*Quaternary Convergence theme (if applicable): *Fundamentals of Health & Disease*Keywords: *osteoarthritis; joints; quality of life; biomechanics; imaging; personalized medicine***1.b Summary (max 300 words)**

Healthy Joints are essential for physical functioning throughout life. Among joint diseases osteoarthritis (OA) has the highest prevalence and disability. In coming decades OA will become the most common chronic disease in the Netherlands.

There are still enormous challenges in OA, ranging from poor understanding about the disease to insufficient early diagnostics and ineffective and non-personalized treatment strategies. To move the field forward, we aim to prevent, diagnose and treat OA at an earlier stage, and target personalized management. For this, it is imperative to improve our understanding of OA phenotypes and how these can be detected early, to develop and apply innovative technology at all anatomical scales and phases of the disease, and to develop and test preventive or early-stage personalized interventions. These efforts will be aligned with the stakeholders' acceptance, and behavioural aspects.

The flagships' researchers with fundamental biological, social, medical and engineering backgrounds will join forces to address: 1) Developing precise personalized computational models for healthy joint growth and maintenance, assess joint load in high-risk and early-stage OA groups, enabling artificial testing of new and precise preventive and early-stage interventions; 2) Build cellular models for different joint tissues and phenotypes to identify personal drivers of OA, and use joint-on-a-chip to test interventions at cell level; 3) Improved and novel diagnostics based on novel hybrid imaging modalities and computational biomechanics enabling the early diagnosis of adverse cartilage load and metabolic joint processes and subsequent targeted therapy; 4)

Machine-learning techniques to better recognize the early imaging joint tissue changes, and the phenotype specific patterns; 5) Test preventive treatment, early-stage diagnostics and treatment combined with behavioural insights and stakeholder choices.

The long-term goal of *Healthy Joints* is to prevent OA where possible, or identify the disease early and precise, to allow early-personalized treatment, yielding at least 5 good quality-of-life years.

1.c Public Summary (max 300 words)

Layman's summary.

Artrose is een omvangrijk probleem, voor de patiënt vanwege een groot verlies aan kwaliteit-van-leven, en voor de samenleving vanwege de verminderde werkcapaciteit en de medische kosten.

Artrose zal in onze samenleving de komende jaren nog aanzienlijk toenemen; binnenkort is het zelfs de meest frequente chronische aandoening. Pijn bij artrose is moeilijk te behandelen, en de ziekte zelf is niet te genezen. Daarom zullen we artrose waar mogelijk moeten voorkomen (preventie), of veel eerder in het ziekteproces én specifiek voor iedere patiënt moeten ingrijpen.

Hiervoor is het essentieel dat we artrose al vroeg herkennen en de specifieke subtypes onderscheiden. Dat gaan we doen door artrose in al zijn facetten te onderzoeken, met gebruikmakend van moderne technologieën om dit zo nauwkeurig mogelijk te doen. Vanaf het moment dat het gewricht zich ontwikkelt op kinderleeftijd, tot het punt dat er vroege artroseverschijnselen zijn. Hiermee zullen we subtype-specifieke en vroege behandelingen ontwikkelen om in de klinische praktijk te worden ingevoerd.

We hebben hiervoor vanuit het Erasmus MC de klinische, radiologische, genetische en laboratorium expertise, en onderzoeksgegevens van grote patiëntverzamelingen beschikbaar. Dit wordt gecombineerd met de data-analytische expertise en de biomechanische expertise van de TU Delft voor de vroegherkenning en patroonherkenning van artrose, het bouwen van simulatiemodellen, en voor precieze en vroege diagnostiek van de gewrichtsbelasting en de ziekteprocessen. Met de expertise vanuit de Erasmus Universiteit gaan we na hoe de voorkeuren en gedragscomponenten van de patiënt een rol spelen bij behandelkeuzes in een vroeg ziektestadium, en wat de kosten en baten op de lange termijn zijn.

Na deze grondige analyses zijn de ontwikkelde persoonsgerichte behandelingen gereed om in patiënten te testen. Hiermee zetten we een belangrijke stap in de preventie, en vroege patiënt-specifieke behandeling van artrose. We streven ernaar om 5 gezonde jaren toe te voegen aan het leven van de artrosepatiënt.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e (now: 2365)

2.a Contribution to H&T mission and fit with H&T theme(s)

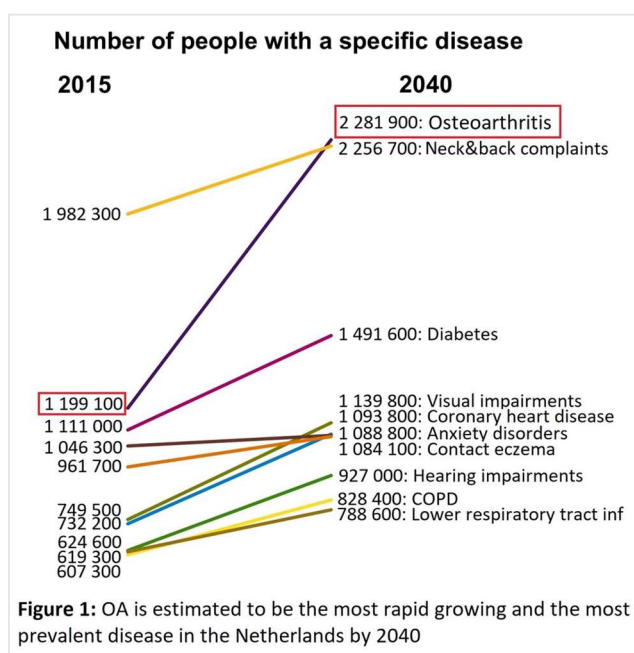
This proposal aims to advance the *Healthy Joints* program to a convergence flagship program, being committed to innovative science targeting to force a paradigm change in the treatment of joint diseases. Starting with lower extremity (hip and knee) osteoarthritis (OA), we aim at long-term to scale up to other joint sites. Given the burden of OA, this will yield a considerable impact on the quality of a (healthier) life, and socioeconomic equality. With early and precise diagnostics as a

starting point, we aim to create successful treatment of OA, individually tailored to the identified aetiology (~OA phenotypes) and subject characteristics/preferences. We even take it further to the development of preventive strategies in clearly defined high-risk groups.

The *Healthy Joints* program is therefore extremely well aligned with the mission of the TUDelft, ErasmusMC/EUR *Convergence Health & Technology Initiative*, that strives to impact healthcare by means of innovative transformations, resulting in improved health of our citizens and their societal participation. The *Healthy Joints* program shares the challenge of the convergence initiative to make this happen without rising healthcare costs.

Before elaborating on how convergence of health and technology will enable such paradigm change and subsequent impact, it is important to shortly review what the foreseen burden of OA is, and why

attempts for a disease modifying treatment are not yet successful.



Osteoarthritis (OA) is already one of the most prevalent diseases and is by far the largest growing disease in the Netherlands. By 2040 OA will be the most common chronic disease in the Netherlands (see Figure 1), with an increase from 1.2 million patients in 2015 to more than 2.2 million in 2040. [1] This concerning trend in the Netherlands reflects the worldwide trend. [2]

The socioeconomic burden of OA is huge. Its pain and mobility limitations have a great impact on the quality of life. Moreover, patients with OA work less and quit their professional life earlier, with an estimated cost of 0.4% of the Gross Domestic Product

(GDP, meaning 3 billion Euros annually in the Netherlands). [3] The total cost (medical and societal costs) is estimated at 1 to 2 % of GDP. [4]

In the current patient journey a whole window of opportunities to treat OA gets lost. After an initial diagnosis of OA (by the general practitioner after pain complaints) the typical treatment is the prescription of pain medication, and unspecified exercise advice or therapy. This is considered picking and keeping up, mainly because no (non-specific) treatment is effective or only modestly effective, and disease-modifying treatment like for rheumatic disease is lacking for osteoarthritis. This lasts until the pain is unbearable, to justify major and costly surgery with a joint arthroplasty. Such an artificial joint has a survival in the body up to 20 years, but is not always possible, and far from successful in every patient. The main problem is that a whole window of opportunities was lost.

The reason why no non-specific treatment is ineffective is because OA is a multifactorial disease to which biochemical, genetic and mechanical factors contribute. [4] The rate of joint degeneration depends highly on the mechanical load. [5,6] This is especially apparent with abnormal joint shape, or changes in the mechanical environment of a joint, e.g. due to injury, which largely increase the risk of developing OA. [5,7,8]. However, current disease modifying interventions are mainly directed at the biochemical disturbances and have not been tested phenotype specific, and not in an early stage. A main reason for no success may be the late irreversible stage of testing and that as long as

biomechanical disturbances are not addressed, important drivers of biochemical disturbances remain. The impact on the joint tissues of such extrinsic biomechanical factors will differ in the presence of certain intrinsic factors (for instance genetic profiles, or immuno-reactive/inflammatory profiles) that make the joint already more vulnerable. A better understanding of the mechanisms of joint (mechanical) homeostasis will enable clinicians to provide personalized guidance and apply personalized therapies, to effectively slow down the (effects of) OA.

The *Healthy Joints* program aims for a paradigm shift in the treatment of OA from a one size fits all approach in patients with established disease towards a person specific treatment (and ultimately prevention) approach in early stage disease.

By recognizing OA disease at an early stage, early stage phenotype specific treatment will become possible. By understanding the multifactorial nature of the disease, we will work towards comprehensive approach that is based on acknowledging different phenotypes. Biological and biomechanical disturbances (pathophysiology and pathomechanics) and their interaction play a role, and different phenotypes of OA have been recognized at group level, but are hard to identify in individual subjects, especially at late stage disease when more joint tissues become affected. In particular precision diagnostics and precision biomechanics are lacking. Although mechanomics is an emerging field in the mechanobiology, this is yet limited to cell level. Moreover the role of radiomics, and genomics is still underexplored. Precision diagnostics will enable early, personalized, targeted therapies, as a real paradigm change to impact the consequences of OA. For clearly defined high-risk groups, preventive medicine can become a real game-changer as well. As joint shape plays such an important role in the development of OA (for example hip dysplasia and femoroacetabular impingement are the main risk factors for hip OA [8,9]), understanding and modelling the joint will greatly add to development of preventive measures during growth as the joint shape after closing of the growth plates at age 13-15 years is final.

While technology is instrumental in the *Healthy Joints* program, we will combine the expertise of different disciplines to establish a real comprehensive approach. This will greatly support the transition of our healthcare system, from reactive (late treatment) to proactive (early or preventive treatment), from general to individual, and from hospital to primary care and in the community

As elaborated in the long-term sustainability plan (2c) the long-term goals of *Healthy Joints* fit the transition theme (to shift care from the hospital to primary care) and Health Journey theme (a shift to early detection and treatment and where possible eventually prevention). However, for the time horizon of this proposal (i.e. the first iteration cycle, see 2c) **the primary fit of *Healthy Joints* program to the themes of the Health and Technology Convergence initiative is to Human Centered Technology and AI for Health**

2.b Synergy between disciplines

The current *Healthy Joints* program is already essentially based on connecting the siloed approaches to osteoarthritis, by creating synergies between epidemiology, orthopaedics, biomechanics and radiology.

In this proposal this approach is pushed further by incorporating more disciplines from the convergence initiative, broadening up to involve more technological groups, i.e. extending the scale of biomechanics, ranging from movement biomechanics to tissue biomechanics to cell biomechanics. We also broaden our approach towards understanding the growth of joint, as an abnormal shaped joint is a major risk for developing OA (already at young adult age). Also new imaging technologies will be explored, aiming to identify new imaging biomarkers of early OA. Moreover data scientist will be involved to apply state of the art machine learning techniques to

process the multimodal diagnostic data into the classification of phenotypes of OA. Finally also the behavioural sciences and health economic science are entering into the multidisciplinary approach of *Healthy Joints*, because understanding of behavioural patterns and preferences is essential to effectively implement the paradigm change that enables early stage diagnostics or even prevention when disease symptoms are limited or absent, behavioural patterns and preferences become increasingly important. This considers trust in counselling health professionals, personal preferences in therapy choice, and therapy compliance.

The multidisciplinary focus of the *Healthy Joints* program is reflected in the composition of the team and the different work packages (see: 3c).

2.c Long term sustainability plan

Sustainability is critical to achieve the goals of *Healthy Joints* since it takes at least a decade, and maybe more, to fully shape and implement the paradigm change in the treatment of OA.

Our vision to constitute a sustainable flagship program is that of repetitive iteration cycles. This means that in each phase a complete Iteration cycle is completed (see figure 2). A more precise representation of the cycle will be found in the work plan.

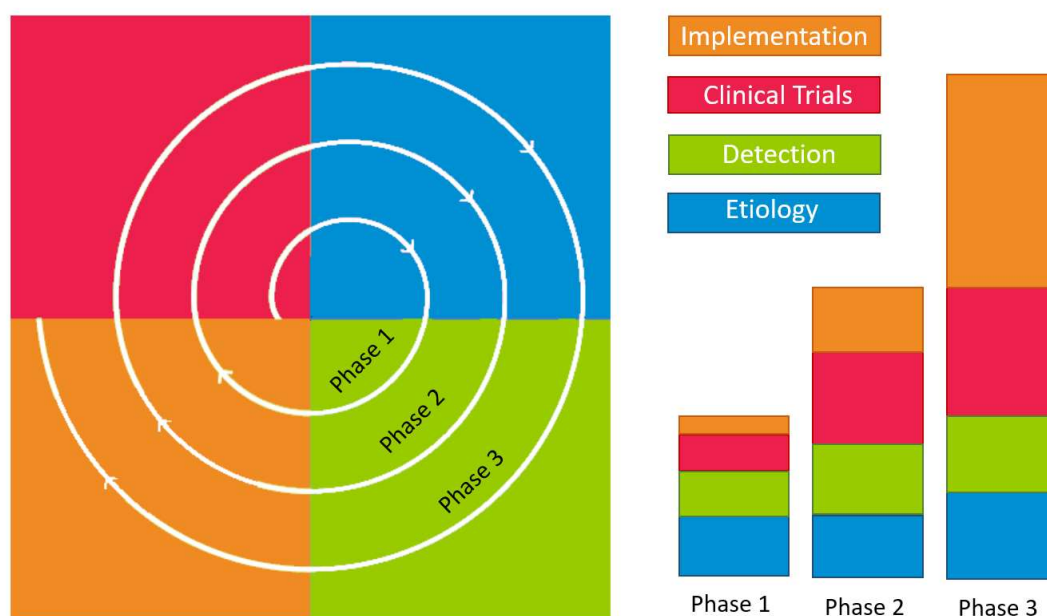


Figure 2. Left (1a): The Iteration approach of system development will be applied to long-term *Healthy Joints* program. Right (1b): In each phase the relative efforts from the respective activities are expected to differ (in a context of overall growth of the program)

Although iteration cycles will be present at multiple time scales, overall we foresee 3 phases: Phase 1 (2022 -2027, i.e. the time window of this application), phase 2 (2027-2032) and phase 3 (2032-2037). Although full societal Impact of the "healthy joints" program will obviously be reached at the longer term, even in phase 1, first steps to perform clinical trials and implementation trials will be executed. Those are based on the current, though yet limited, insights of specific aetiology and targeted treatments in OA. In the larger context these will inform the next phases. Altogether, **the main focus of phase 1 is to unravel relevant aetiology, using technological innovations, to enable to detect OA and intervene at a very**

early stage, or even in a preventive manner. This will inform future clinical trials in all their phases and those outcomes will inform future implementation trials that eventually will yield full societal impact.

While we expect the absolute amount of the *Healthy Joints* program **will double every phase**, we also foresee that the relative amounts will differ to reflect what is needed in the different phases of the *Healthy Joints* program, see figure 1b.

The scientific Impact of *Healthy Joints* will also be considerable since unravelling the aetiology of OA, albeit for now (just) to the extent that it serves early detection and phenotyping of OA, and understanding childhood joint growth, will require major scientific breakthroughs. These will appear in the setup of biomechanical multiscale modelling, joint growth modelling, new imaging techniques and mechanobiological discoveries. While in the above the pathway to societal impact has been described, these scientific breakthroughs will also foster the development of new therapies (e.g. based on tissue engineering of cartilage) in OA or to prevent OA, while the methodological approach can be applied to other bone and joint diseases. Moreover in the behavioural area, insights in patient- and healthcare providers preferences, and therapy compliance will be generalizable to other areas.

Risks: the main risks of the healthy joints program are twofold. First risk is that the scientific breakthroughs that we aim for will not happen. While this is an obvious risk of every scientific program, the extent of this risk seems limited. Many of the research questions that will be addressed in the WPs are based on previous research of ourselves and others, keeping feasibility within the equation. The unique proposition in general is that the innovation is driven by multidisciplinary collaboration, not a high-risk mono-disciplinary aim.

The second risk is that the multidisciplinary collaboration will fail, meaning the *Healthy Joints* program falls apart. This seems unlikely, since there already is an established collaboration between the main leads and many of the *Healthy Joints* program members. With a track record of collaborative efforts and the context of the strategic CHT initiative, any problems with respect to collaboration are likely to be solved.

2.d Visibility, attracting talents and new partners

By nature of the research program there will be a lot of opportunities in the projects that constitute the *Healthy Joints* program. Actually it is not just an opportunity but also a must for a successful program. By nature of the research activities of *Flagship* members (who participate in the university educational program anyway) there is an intrinsic drive to include both bachelor and master students in their projects. The project co-lead prof Harlaar is educational director of university institutional programs Clinical Technology (Bachelor) and Technical Medicine (3 year master), a joint degree between Erasmus en TUDelft that integrates engineering and medical knowledge, skills and mindset towards a new professional.

The *Healthy Joints* program has already gained considerable exposure. Sita Bierma-Zeinstra is listed as the worldwide no1 hip OA researcher, a position that attracts PhD students and early career researchers to work for the program under her lead. Jaap Harlaar and Edwin Oei were each invited as keynote lecture of the prestigious "Year in Review mechanics" and "Year In Review Imaging" lectures at the OARSI congress 2021, that provided ErasmusMC/TUDelft with quite some visibility as an Innovative research consortium for OA. Also in the national press (Sita Bierma-Zeinstra pledging for an "Osteoarthritis offensive ") and finally the Biomechanics in OA project was selected as showcase to all Delft University of Technology alumni.

The *Healthy Joints* program already makes active efforts to attract new early career researchers by taking the lead in a Marie Curie ITN proposal to be submitted In 2022, as a follow up of the EU KneeMo ITN-network, by jointly employing 10 Early Career Researchers on OA research.

2.e Long term societal impact

For the actual Impact on the long term, the *Healthy Joints* program's goals aligns with the Dutch national goals for healthcare in 2040, i.e. to add 5 healthy years to people's lives, and to minimize societal health inequalities. As can be concluded from 2c (i.e. the long term sustainability plan), implementation will become a greater part of the next phases in *Healthy Joints*. In the next phase also other societal stakeholders will be involved, and cost effectiveness studies of the new approach. Although the societal Impact is at somewhat longer term, phase 1 of the *Healthy Joints* program already includes some first steps on this path.

OA is predicted to become the most prevalent chronic disease in the Netherlands and confers a personal burden, manifesting as low physical activity, elevated body mass index (BMI), social disengagement, psychological distress, and poor quality of life. [9] It increases the risk of chronic diseases such as heart disease and diabetes and cardiovascular-related death. [10] The combined annual healthcare and lost productivity costs of OA in the Netherlands are €3 billion annually. Preventing hip OA has never been more important.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d (now: 1998)

3.a Ambitions

Despite the impact of the most prevalent joint disease osteoarthritis (OA), there is no disease modifying treatment or cure; at best patients receive modestly effective symptomatic treatments, while the success of expensive joint replacement in end-stage OA is limited. Given the slowly progressive nature of the disease, with irreversible structural change leading to chronic pain and disability, early pro-active interventions or even prevention would be the most logical treatment strategy.

The ambition of the *Healthy Joints* flagship is to advance the research field of joint health. We do this by refocusing the research field of osteoarthritis (OA) towards prevention and very early stage personalized treatment.

The challenges that arise are first the scientific ones: because in order to enable very early stage personalized treatment the aetiology of the complexity of OA needs to be unravelled and made applicable to the individual level, in a clinical context. The second main challenge towards very early treatment and prevention is with respect to its future implementation, taking behavioural changes and stakeholders attitudes preferences into account.

Innovations in this program are early OA detection, early stage OA phenotyping, modelling healthy joint growth, and modelling healthy joint load, development and testing of treatment in early OA and preventive treatments in high-risk groups.

Developing and applying new technologies, i.e. biomechanical multiscale analyses, new imaging modalities, advanced computational models and data analytics, and mechanobiological modelling at cell level will be the key for a breakthrough in the personalized early or preventive treatments, and can only be achieved by combining expertise from biomedical and technical disciplines. Along this process, preferences and acceptance of stakeholders, and facilitating behavioural changes towards healthy joint load, will be directing the clinical application.

3.b Aims and feasibility

The objective of the *Healthy Joints* flagship is to constitute a comprehensive research program to advance prevention and early diagnostics & treatment for OA, starting with knee and hip OA the first 5 years but anticipating to expanding to other OA joint sites.

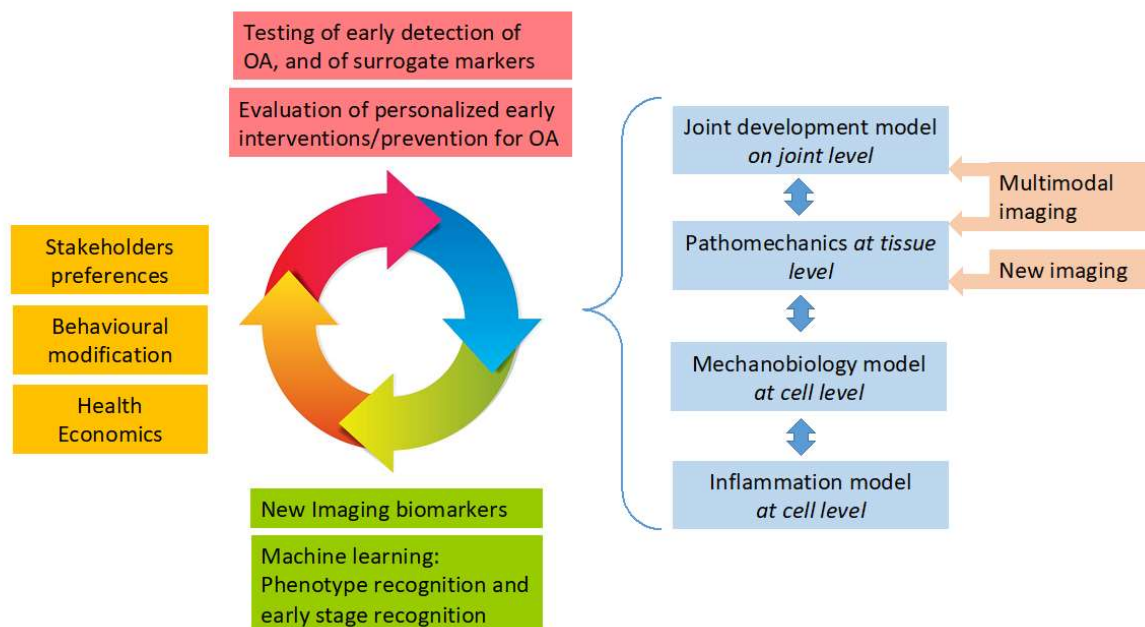
The Feasibility of *Healthy Joints* is excellent. Erasmus MC is already worldwide at the forefront of OA research in defining early-stage OA, and testing early-stage and preventive interventions.

The flagship *Healthy Joints* builds further on already existing collaborations between the different universities and parts of this program already began to run based on recent grants (of which 2 EMC/TU Delft convergence grants). We have the clinical, technical, social, and biological expertise, the technologies (ready or in development), the running and pending research grants, and the multiple existing datasets in place. Many of our flagship members already operate in the world top of their research field.

3.c Work plan

Our proposed flagship *Healthy Joints* is a fundament on which we will expand our united efforts to enable early stage (personalized) treatment and prevention for OA. This *Flagship* defines our directions of research and grant proposals, will strengthen our interdisciplinary interplay, and enables our next generation researchers to grow and excel.

Figure 3: The *Healthy Joints* program: in blue WP2-4; in green WP5-6; in orange WP7; in red WP8



In this flagship the different pillars of research to accomplish our main objective are reflected in the different work packages (WPs). The WPs have a combination of a lead and co-lead that are a senior established researcher and an upcoming high potential researcher. We provide a constant iteration between the WPs in order to guarantee a continuous and integrated progress towards our main objective. Below you find a short description of the different WPs. **The mutual connections are depicted in Figure 4.** The deliverables that will be achieved with already existing grants (during the

first years of the program) are marked with an asterisk(*). **WPs 2-8 also all have as deliverables to submit further grant proposals for their WPs.**

WP 1: Governance and Innovation management of the flagship

Deliverables: 1) Monthly short symposia for all members, PhD students and master students; 2) Twice a year a flagship meeting for all flagship members and involved PhD students; 3) Quarterly WP leader meetings; 4) A grant proposal plan; 5) A career plan for all post-docs and mid-career researchers involved; 6) Valorisation plan and stakeholders involvement plan; 7) Deliver one year reports on program progress.

Members: Lead: Bierma-Zeinstra(EMC), co-lead: Agricola(EMC); members: Merkelbach(EUR), Harlaar(TUD).

WP 2: Modeling of healthy joint growth (joint level)

Deliverables: 1)*^{partly} world-wide unique 2D and 3D joint shape and growth data of hips and knees in children and adolescents; 2)* identified high-risk groups for developing abnormal joint shape 2) A novel 5-D hip growth model including 3D shape, density and time; 3): a computational biomechanics-modelling platform for testing mechanisms and potential interventions during childhood; 4) Potential remedial options for dysplastic or CAM hip growth.

Facilities & infrastructure: 2D (semi)automatic statistical shape modelling program for hip DXAs, and 3D segmentation of MRIs are already in place. Generation R cohort data: multimodal imaging (hip DXAs, hip MRIs, knee MRIs, whole body DXAs, and whole body MRIs) during the whole childhood until 13 years already in place (imaging available in 6000 children in the Generation R cohort). At age 18 the measurements are ongoing. A wealth of determinants during foetal life, infancy and the whole childhood already available in the Generation R cohort.

1st Iteration: Potential interventions during growth to turn dysplastic growth to normal development of the hip, or to prevent femoroacetabular impingement.

Members: Lead: Zadpoor(TUD), co-lead Sarkalkan-Tumer(TUD); members: Bierma-Zeinstra(EMC), Agricola(EMC), Oei(EMC), vMiddelkoop(EMC), Klein(EMC), Tolk(EMC).

Funding: A ReumaNederland pilot study (Bierma-Zeinstra) is running for 2D hip growth data, a PHD trajectory is almost finished for 3D hip segmentation (Oei), a ReumaNederland pilot study (van Middelkoop) for 3D segmentation of the knee just started. A CSC PhD student (Agricola/Bierma-Zeinstra) for femoroacetabular impingement in generation R recently started.

Grants pending: ERC Advanced "HIPSTAR" (Bierma-Zeinstra)

WP 3: Multiscale modeling pathomechanics of joint and tissue load In vivo (tissue level)

Deliverables: 1)* precise quantification of gross joint load, upon different functional tasks; 2)* precise quantification of resulting cartilage stress using personalized models; 3)* evaluation of cartilage stress in specific high-risk groups of OA, and in different early OA phenotypes.

Facilities & Infrastructure: 3D motion capture to measure full body motion on an instrumented treadmill to measure external forces, combined with electromyography (EMG), and biplane video-fluoroscopy to collect radiographic images using two synchronized X-ray tubes during motion is in the set-up phase. The designated location in EMC is being rebuild for this purpose. PET-MRI facility at EMC already operational.

1st iteration: To develop a multiscale (whole body + tissue) pipeline by integration of Imaging modalities and biomechanical model (whole body + tissue based data processing, develop a first generation (candidate set) of pathomechanics/degeneration.

Members: lead: Harlaar(TUD), co-lead: Wesseling(TUD); members: Zadpoor(TUD), Seth(TUD), Macri(EMC), Oei(EMC), Bierma-Zeinstra(EMC), van Middelkoop(EMC), Meuffels(EMC).

Funding: Convergence project (Harlaar/Bierma-Zeinstra), ZonMW-Open project (Bierma-Zeinstra/Harlaar/Oei) including 250K investment.

Grants pending: VIDI (Wesseling), NWA-ORC (Harlaar/van Middelkoop)

WP 4: Mechanobiological and inflammation cell models of OA development (cell level)

Deliverables: 1) Insight in the effect of different mechanical stimuli on cartilage maintenance and degeneration obtained with cells and tissues in vitro considering the influence of degenerative cartilage changes (incl. local heterogeneity in matrix); 2) New in vitro models representing the effect of different types of inflammation (e.g. different cytokines, synovium, macrophages, senescence) and genetic variation that include mechanical loading to study disease mechanisms; 3) Different personalised pharmaceutical interventions (linking Intervention to disease phenotype) tested in the new in vitro models and selected for translational studies.

Facilities & Infrastructure: nanoscale mechanical testing infrastructure TUD, Optical Imaging Center, IPS facility, Genomics facility, laboratory facilities present; use of human tissue from total knee replacement surgery.

1st iteration: in vitro (at tissue and cell level) mechanobiological models, including the role of Inflammation and genetic background. Next iteration: refine the models by including bone; use the models to develop/test new treatments; translate/scale findings from cell to the patient (connect with WP3)

Members: Lead: vOsch(EMC), co-lead: Narcisi(EMC); members: vMeurs(EMC), Boer(EMC), Lubberts(EMC), Koenderink (TUD), Mirzaali(TUD).

Grants pending: NWA-ORC (vOsch), and NWO/TTW perspectief (vOsch)

WP 5: Multimodal machine learning approach to identify early stage OA phenotypes and features

Deliverables: 1)* Machine learning methodology that can deal with missing data and integrate different cohorts; 2) novel machine learning algorithms to identify early-stage OA phenotypes and features; 3) Early-stage OA phenotypes and OA features

Facilities & Infrastructure: GPU and CPU clusters and XNAT platform at EMC enabling large-scale automated multimodal data processing; Existing data sources by means of relevant and deeply phenotyped patient or high risk-cohorts: Rotterdam-Study (n~15000), LITE-study (n=234), PROOF-study (n=400), CHECK-cohort (n=1002), OAI-cohort (n=4492), KNALL-cohort (n=154), MOST-cohort (n=3021), World-COACH dataset (individual patient data from all hip cohorts worldwide; n=32000), Generation-R cohort at adult age (n=4000).

1st iteration: To use multimodality data from existing databases and clinical trials to predict development of OA and uncover different progression phenotypes. Next Iteration: Incorporate results from the previous round with new data modalities (e.g., PET-MRI, photon-counting CT).

Members: Lead: Krijthe(TUD), co-lead: Reinders(TUD), Hirvasniemi(EMC), Agricola(EMC), Oei(EMC), Klein(EMC), Loog(TUD), vMeurs(EMC), Boer(EMC), Bierma-Zeinstra(EMC), Runhaar(EMC), vMiddelkoop (EMC), Meuffels(EMC).

Funding: Convergence project (Klein/van Meurs/Loog/Krijthe); Erasmus fellowship (Agricola), ReumaNederland (Agricola)

Grants pending: NWO Open-M (in preparation, Krijthe/Klein)

WP 6: New imaging techniques and imaging biomarkers to assess the early OA processes

Deliverables: 1) novel, validated, quantitative imaging biomarkers for early OA based on (PET-)MRI, photon counting CT (PCCT) and ultrasound, including those suitable as new surrogate outcomes; 2) new, automated (artificial Intelligence based), image analysis algorithms for tissue segmentation, quantitative biomarker extraction and radiomics.

Facilities & Infrastructure: All high-end equipment, Including PET-MRI and PCCT, is already operational at EMC.

1st iteration: Accelerated acquisition of validated imaging biomarkers, particularly in MRI, to enable fast and feasible imaging protocols for widespread use.

Members: Lead: Oei(EMC), co-lead: vdHeijden(EMC); members: Renaud(TUD), Weingartner (TUD), Klein(EMC), Hirvasniemi(EMC).

Funding: PhD and post-doc projects (Oei) including Initial studies of knee OA on PET-MRI and photon counting CT already Initiated. ZonMW-open (Bierma/Harlaar/Oei)

Grants pending: ERC consolidator (Renaud)

WP 7 Behavioral attitude characterization, and the role of attributes towards stakeholder choices and preferences.

Deliverables: 1) behavioral components and determinants of treatment seeking, and treatment compliance; 2) intervention options supporting treatment seeking and treatment compliance; 3) contribution of the different attributes to the willingness to undergo or apply certain preventive or early-stage treatment.

Facilities & Infrastructure: Artrose Gezond patient platform and panels. HEALTHY'R cooperation with the municipality of Rotterdam. Academic (research) network of primary care practices (PRIMEUR).

1st iteration: behavioral analysis of determinants of early health consultation, therapy compliance; patient Involvement artrose gezond). Next Iteration: development of behavioral Interventions, using behavioural Insights and co-creation, health-economics.

Members: lead: Merkelbach(EUR), co-lead: de Bekker-Grob(EUR); members: Schiphof(EMC), Bos(EMC), Damen(EMC), Tolk(EMC).

Funding: Business case Artrose Gezond (patient platform)

Grants pending: Stichting Theia implementation project (for patients with low literacy)

WP 8 Testing early detection and interventions in clinical populations

Deliverables: 1)*partly feasibility; 2) efficacy 3) effectiveness 4)*partly efficiency of early-stage and phenotype specific detection and interventions; 5) validation of surrogate outcomes in preventive and early stage treatments; 6) feasibility and efficacy of preventive interventions

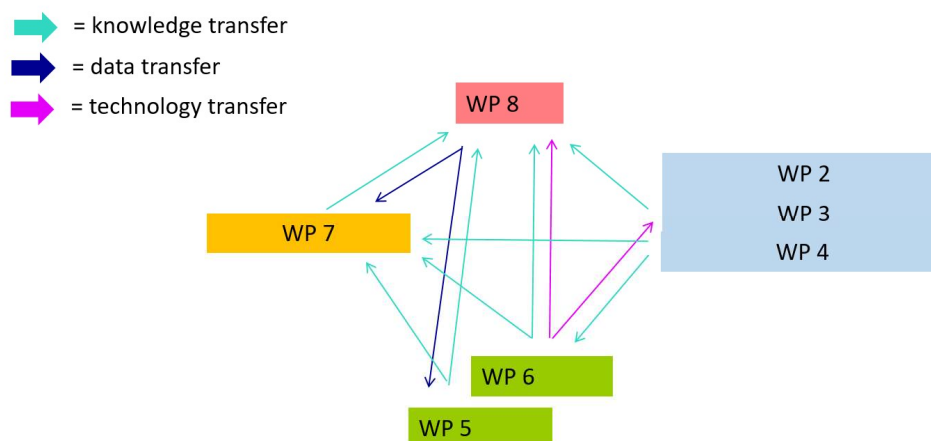
Facilities & Infrastructure: Trial unit at the department of General Practice; Academic (research) network of primary care practices (PRIMEUR) providing care for inhabitants of the Rijnmond region.

1st Iteration: at least pilots/RCTs on exercise, GLI and orthoses, POC trials on gait retraining; Next Iteration: nation-wide studies using the results of the first Iteration, health economic modeling.

Members: Lead: vMiddelkoop(EMC), co-lead: Runhaar(EMC); Bierma-Zeinstra(EMC), Meuffels(EMC), Bos(EMC), Tolk(EMC), Schiphof(EMC), Merkelbach(EMC), de Bekker-Grob(EMC).

Funding: ZonMW Pilot trial on exercise in early stage knee OA (Schiphof), ZonMW cost-effectiveness trial on combined life-style (GLI) intervention (Middelkoop) in early stage knee OA.

Grants pending: ZonMW Proof of Concept study (Bierma-Zeinstra/vMeurs), ZonMW hulpmiddelen pilot study (vMiddelkoop); VIDI on surrogate marker for early OA (Runhaar).

Figure 4: *Mutual connections of WPs:*

3.d Alignment of the program with relevant national and international research agendas

Healthy Joints aligns perfectly with the national “Artrose Alliantie” by ReumaNederland (Dutch Arthritis Association); the impact goals set by the Top Sector Life Sciences & Health (Health~Holland): *5 more healthy years by 2040* ; the Nationale Wetenschaps Agenda routes: *personalized health care, regenerative medicine*, and the European Joint Action on Chronic Diseases and Promoting Healthy Ageing across the Life Cycle (JA-CHRODIS); The National Prevention Programme; National agenda Sports and Physical Activity.

4. Consortium (25%) max 2000 words for 4.a to 4.d (now 1995)

4.a Track record of the consortium members

The flagship program *Healthy joints* is proposed by a team of 31 scientists from 12 departments (6 ErasmusMC, 2 EUR and 4 TUDelft). This consortium is composed from a range of disciplines but also with diversity in seniority (also with regard to (co-)leads) to warrant its viability. Below we just mention some track record of the leads (and some others).

WP 1 flagship governance is led by Prof. Bierma-Zeinstra (WSc H-index 74; 500 international indexed papers), she is also the main lead of the flagship. She attracted multiple grants (14 million as PI, and 12 million as co-applicant) with prestigious grants like VIDI, ZONMW-open etc. She leads the international OA Trial Bank with IPD of >15000 OA patients in RCTs. She was awarded by the Osteoarthritis Research Society International (OARSI.org) with the clinical research award to recognize excellence in research related to OA over the past five years. She also was supervisor of the year (chosen by the more than 1000 PhD students of EMC). The co-lead of WP1 is dr. Agricola (WSc H-index: 24, 73 papers) is a high-potential early career researcher and orthopedic surgeon. He leads the world-Coach consortium (IPD from > 40.000 hip OA patients). He already received a VENI grant, a prestigious ErasmusMC fellowship and two ReumaNederland grants. Agricola initiated and steers the world COACH consortium. Dr. Merkelbach (EUR), co-lead of the flagship program is a young career researcher at the EUR with 6 papers; she received a ZonMW Covid grant and is managing director of Behavioural Insights Group Rotterdam (BIG'R). Prof. Harlaar (TUD) also co-lead of the program (WSc H-index 39, 219 papers) received over the last 10 years 2.2 million as PI and 18 million as co-PI (of which H2020 MSCA-ITN consortia, and multiple STW programs). Harlaar designed various technical modalities of biomechanics based clinical movement analyses that were valorized and broadly introduced in clinical practice, amongst which the GRAIL, now a product of MOTEK medical. In march 2020 (at the onset of the COVID 19 crisis) prof. Harlaar was leading a group of 30+ students, who developed an IC ventilator within 3 weeks from scratch, 80 pieces were bought as backup for the Dutch healthcare system. Harlaar is also is the Educational Director Clinical Technology & Technical Medicine, and was co-lead in several large EU consortia and past president of international scientific societies.

The **WP 2** leader Prof. dr. Amir Zadpoor (WSc H-index 55, 274 papers) has received : an ERC grant, a Veni grant, and the Early Career Award. The co-leader, Dr. Sarkalkan-Tumer (WSc H-index 9, 15 papers), is an early career researcher. Both take up many MSc and BSc students of biomedical engineering (BME) and technical medicine (TM).

The lead in **WP 3** is done by Prof. Harlaar (see above) and co-lead is dr. Wesseling (WSc H-index 16, 48 papers), and Dr. Seth (24 papers, WSc H-index 18). Dr. Wesseling also worked in musculoskeletal image processing industry (Materialize) while and dr. Seth, who came over from Stanford, is the software architecture of OpenSIM, the de facto standard in scientific musculoskeletal biocomputational simulations. Together they also take up many Msc and BSc students, both TM and BME.

Prof. van Osch (WSc H-index 54, 240 papers), **WP 4** leader, has received multiple large grants of which two H2020 MSCA-ITN consortia, and STW grants, and served in steering committees of multicenter European and national projects. Last year she was awarded with the prestigious 'Engineering and Physical Sciences' Suffrage Science award. The co-leader, Dr. Narcici (WSc H-index 15, 40 papers) holds ReumaNederland grants and an Erasmus MC grant. Also Assoc prof Lubberts (WSc H-index 54, 154 papers) cooperates in this WP. Prof. van Meurs (WSc H-index 71, 243 papers) received 2 million in grants the last 5 years, and cooperates in the world-wide Genetics of Osteoarthritis consortium, and NIH grants. These members supervise many MSc students from the life sciences.

From TUD also Prof. dr Koenderink (WSc H-index 45, 140 papers) is involved. She received various distinctions, including an NWO VIDI (2008), elected membership of the Young Academy of the KNAW (2008), ERC Starting Grant (2013), NWO VICI (2019) and the P-G. de Gennes Prize (2018), and she is a co-recipient of the NWO-Gravitation program Basyc (2017-2027).

The **WP 5** leader, Dr. Krijthe (28 papers, WSc H-index 7), has published theoretical work in leading AI venues and was co-applicant of the NWO-Crossover MOCIA project. Co-leader Dr. Loog (159 papers, WSc H-index 22) is an expert in supervised machine learning and is deeply involved in TUDelft's machine learning education, with extensive experience involving MSc students in research and is co-PI for one of the first Delft AI labs. Co-lead Prof. dr. Reinders (303 papers, WSc H-index 48) is an expert in bioinformatics and machine learning and holds multiple grants from ZonMW (VOILA), NWO (Person2Data) and Health-Holland. Their counterparts at EMC are Dr. Hirvasniemi (WSc H-index 9, 30 papers), who has received a LEaDing Fellows postdoctoral fellowship (Horizon2020 Marie Skłodowska-Curie COFUND programme), postdoctoral fellowship, dr. Klein (Google H-index 45, 98 papers) is co-applicant and WP-leader on multiple H2020 and NWO projects, and prof dr. van Meurs (WSc H-index 71, 243 publications) is the (co-)applicant of multiple ZonMW and NIH related to OA.

The **WP 6** leader, Assoc prof Dr. Oei (WSc H-index 35, 183 papers) leads the musculoskeletal radiology clinical section and research group at ErasmusMC. He received 2 million in grants as PI, among other from ReumaNederland and the Radiological Society of North America. He was awarded a research fellowship at Stanford University. The co-lead Dr. van der Heijden received a grant from the European Society of Musculoskeletal Radiology and was awarded the departmental Imaging Program of Excellence. She currently is a visiting assistant professor at the University of Wisconsin. The counterpart of the leads at TUD is Dr. Weingartner (WSc H-index 17, 59 papers). He received the prestigious START-UP Grant awarded by the Dutch research Council (NWO). Dr. Renaud (51 papers, WSc H-index 19) from TUD brings the specific expertise on ultrasonic visualization of bone vascularization.

Along with the lead dr. Merkelbach (EUR), Prof. dr. de Bekker-Grob (95, WSc H-index 27) co-leads the **WP 7**, and received many grants including prestigious VENI and VIDI grants, and is director of the Erasmus Choice Modelling Centre. Dr. Tolk (14 papers, WSc H-index 7), pediatric orthopedic surgeon holding two research grants in pediatric surgery, and dr. Bos (66 papers, WSc H-index 25, orthopedic surgeon and leads the orthopedic specialist training, and dr. Damen (WSc h-index: 11, 19 papers), General Practitioner, are the clinicians involved in this WP, accompanied by Dr. Schiphof (36 papers, WSc H-index 15); she hold several grants (> 1 million), mainly ZonMW grants.

The **WP 8** leader, Assoc prof dr. van Middelkoop (WSc H-index 36, 180 papers) has a background in human movement sciences and received competitive national grant support of >EUR 3 million. She was twice awarded with a EUR fellowship (2011 and 2015). The co-lead dr. Runhaar (78 papers, WSc H-index 20), received multiple grants (1.6 million as main applicant, and 3 million as co-applicant), led the first preventive OA trial ever (ZonMW prevention grant), and is now co-applicant of a large NIH preventive OA trial.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Each workpackage a) has aims that are directly derived from the design approach (fig 1 and 2) to achieve the overall goal, i.e. early recognition and personalized treatment (see part 2 and 3), and b) is thoughtfully constructed to boost the multidisciplinary approach that is enabled by convergence of health and technology in which the complementarity of the team members acts as multiplication rather than addition, in short

In **WP 2** Healthy Joints will merge the expertise of *tissue biomechanics* to develop computational growth models with *advanced image processing applied* to a unique data set, governed by an

epidemiological overview. Only this combination of expertise will yield accurate growth and risk models.

In **WP 3** *musculoskeletal biomechanics* (experimental and computational) will be combined with a unique *high precision (biplanar fluor) image acquisition and processing*, together with models of *tissue biomechanics*. Only through this combination the precise individualized cartilage stresses can be obtained, to inform targeted therapies.

In **WP 4** the step from tissue to cell level involves the complementary of the *nanoscale mechanics* of the cell environment, the *biological responses to inflammation*, and its interaction with mechanical stress, together with *cell genomics*. Only taking this range of expertise into account the complex distinction between physiological cell-mechanics and its pathomechanics can be unraveled.

In **WP 5** the combined use of *genetic, clinical, and imaging data*, from combined long term cohort studies, with advanced techniques advanced *machine (deep) learning techniques* brings together not only that unique set of data but also the experts behind that data, to yield meaningful early recognition of clinical phenotypes of OA.

In **WP 6** the search for a new radiologic biomarkers is based on combining *innovative imaging physics* and *image processing (including machine learning)* together with *clinical radiology*.

In **WP 7** the combination of *behavioral sciences, choice modeling, and health economic modeling* together with stakeholders (at least patients and health care providers) is an unprecedented combination of expertise, will yield very important knowledge on how mechanistic insights in the early or preventive treatment of OA may eventually reach clinical implementation.

In **WP 8** the *epidemiological/clinical trial* and *health-economic* expertise's together with *clinicians, patients and implementation expertise* will provide not only knowledge of efficacy and (cost)effectiveness, but also on implementation readiness.

4.c Involvement of partners

We will make use of our patient panels of Artrose Gezond (regional) and of ReumaNederland (national). OA patients are involved in discussions about OA research in all stages from research ideas to communication of results. Furthermore, we consult with a regional panel of general practitioners, and use our collaboration with University of Applied Sciences Rotterdam for research and implementation in physical therapy practices. For new imaging there is a longstanding collaboration with General Electric Healthcare, and for motion capturing with MOTTEK. Existing close collaboration with the municipality of Rotterdam by means of HEALTHY'R helps to tackle behavioral issues related to health, such as for instance promoting physical exercise.

See further the participation board (4.d).

4.d Governance, organization and collaboration strategy of the consortium

The governance of "Healthy Joints" is contained in work package 1. Having a dedicated work package warrants that the flagship leads take the explicit responsibility to oversee the joint development, the synergies of expertise, and options for new opportunities. Also project-progress and realization of funding that matches the input from the convergence initiative will be overseen by the flagship leads. The governance leads are very well embedded in the ecosystems of the three institutes. The senior partners in this Governance WP all have extensive experience in leadership. Project leadership meetings with all WP leaders and co-leaders will take place every 3 months, to discuss progress, issues and objectives (short and long term).

At the level of flagship lead, staff members, postdocs, PhD-, Master- and bachelor-students, every month a *Healthy Joints* symposium will be organized, to report progress, and discuss issues in a safe

environment. During the twice-yearly larger science symposia interactive and exchange methods will be used to exchange expertise and facilitate collaborations within the flagship.

Finally a participation board will be installed, that serves not only as an advisory board, but will be involved with active participation (hence its name) and co-creation. All relevant stakeholders from the Dutch ecosystem will be asked to contribute: patients (i.e. ReumaNederland/Artrose Gezond), companies (current collaborators are MOTEKmedical, MoveShelf, Xsens, Elictac, Siemens), healthcare providers (physiotherapists, GPs and Orthopedic surgeons), governmental policy makers (i.e. Topsector Life Science & Health aka Health~Holland) and healthcare insurance companies. This grenium will embody the Dutch strategy (called *quadriple helix*) to health care innovation. The explicit goal is to initiate projects of co-creation from the Healthy Joints program, even before the start of phase 2.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

IV References

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3. Laires PA, et al. The impact of osteoarthritis on early exit from work: results from a population-based study. *BMC Public Health*. 2018 Apr 11;18(1):472.
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6. Andriacchi TP, et al. A framework for the in vivo pathomechanics of osteoarthritis at the knee. *Ann Biomed Eng*. 2004;32(3):447-457.
7. Simon D, et al. The Relationship between Anterior Cruciate Ligament Injury and Osteoarthritis of the Knee. *Adv Orthop*. 2015;2015.
8. Saberi Hosnijeh et al. Cam Deformity and Acetabular Dysplasia as Risk Factors for Hip Osteoarthritis. *Arthritis & Rheumatology*. 2017;69(1):86-93.
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Budget form: Health & Technology Convergence Flagship Call

Flagship:	Healthy Joints
Flagship acronym:	Healthy Joints
Main lead	Sita Bierma-Zeinstra
period :	1 oktober 2022 - 1 oktober 2027
Totaal project budget:	14.131 kEuro

Base Annual Costs Personnel

114	impl.fellow/year
101	postdoc/year
86	PhD/year

Type

acquired	: project running at the time of application
pending	: pending application at the time of application
aimed for	: new applications to be doe by teh flagship post docs

Date 15 mar 2022

				Year1			Year2			Year3		Year4		Year5		Total	Acquired & Pending Funding References
All numbers in kEuro	type *)	base	fte	Flagship	External	Pending	Flagship	External	Pending	Flagship	External	Flagship	External	Flagship	External		
Workpackage 1																	
Algemene kosten	Flagship	10		10			10			10		10		10			
Workpackage 2																	
ReumaNL PhD	acquired	86	1		86												ReumaNederland
ReumaNL PhD	acquired	86	1		86			86									ReumaNederland
CSC PhD	acquired	50	1		50			50			50		50				CSC - Chinese scholarship
PhDs	pending	86	5			430			430		430		430				ERC-advanced Sita Bierma
Postdocs	pending	101	0,8			81			81		81		81			81	ERC-advanced Sita Bierma
PhD	pending	86															ERC starting Agricola
PhD student	aimed for	86	1						86		86		86			86	
PostDoc	Flagship	101	0,45											45			
Workpackage 3																	
PostDoc	acquired	101	1		101												Medical Delta - IMT
PostDoc	acquired	101	1		101			101			101		101				ZonMw-Open (TOP)
PostDoc	pending	101	1						101		101		101			101	VIDI Wesseling
PhD student	pending	86	1			86			86		86		86				VIDI Wesseling
PhD student	pending	86	1			86			86		86		86				TTW Harlaar/Oei/Wesseling
PhD student	pending	86	1			86			86		86		86				NWA - Load
PhD student	pending	86	1						86		86		86				ITN- Macri
Investering BiFluoLab	acquired				250												ZonMw-Open (TOP)
Investering BiFluoLab	acquired				250												TUdelft afd biomechEng
Investering BiFluoLab	Flagship			75													
Investering BiFluoLab	acquired				75												Orthopeadics/Radiology/UnivFund TUD
PhD student	aimed for	86	1						86		86		86			86	
PostDoc	Flagship	101	0,45											45			
Workpackage 4																	
PostDoc	Flagship	101	0,8	81			81			81		81					
PhD student	pending	86	1			86			86		86		86				NWA - Load
PhD student	pending	86	1			86			86		86		86				NWO perspectief
PhD student	aimed for	86	1						86		86		86				
PhD student	aimed for	86	1								86		86				
PostDoc	Flagship	101	0,45											45			
Workpackage 5																	
PostDoc	Flagship	101	0,8	81			81			81		81					
PhD student	acquired	86	1		86			86			86						ErasmusMC fellowship
PhD student	acquired	86	1			86											ReumaNederland
PhD student	pending	86	1						86		86		86				ZonMw - Vital OA
PhD student	aimed for	86	1						86		86		86			86	
PhD student	aimed for	86	1								86		86			86	
PostDoc	Flagship	101	0,45											45			
Workpackage 6																	
PostDoc	Flagship	101	0,7	71			71			71		71					
PostDoc	pending	101	1			101			101		101		101			101	ERC consolidator - Reaud

PhD student	oending	86	1			86			86	86	86	86	86	ERC consolidator - Reaud
PhD student	aimed for	86	1					86		86		86	86	
PhD student	aimed for	86	1							86		86	86	
PostDoc	Flagship	101	0,5									51		
Workpackage 7														ZonMw stichting Theia
PostDoc	Flagship	101	0,35	35			35			35		35		
PostDoc	Flagship	101	0,4	40			40			40		40		
Implmentation Fellow	Acquired	114	0,13		15			15						
PostDoc	pending	101	1			101			101		101			
PhD student	aimed for	86	1						86		86		86	
PhD student	aimed for	86	1								86		86	
Workpackage 8														ZonMW ZonMW VIDI Runhaar VIDI Runhaar
Flagship postdoc	Flagship	101	0,8				81			81		81		
PhD student	acquired	86	1		86			86		86				
PhD student	acquired	86	1		86			86						
PostDoc	pending	101	1			101								
PhD student	pending	86	1			86			86		86			
PhD student	aimed for	86	1						86		86		86	
PhD student	aimed for	86	1								86		86	
PostDoc	Flagship	101	0,5									51		
Total Flagship application				393			399			399		399		1988
Total external input					1272	1502		510	2018		2771		2670	1401
Matching factor per year, first two years, only acquired funding					3,2			1,3						
Matching factor per year, all years *) : acquired, pending and aimed for						7,1			6,3		6,9		6,7	3,5
*) NB. it must be realized that this matching is dependent on the succes rate of both the pending as well as the 12 PhD postions to be applied for by the new flagship postdocs ("aimed for"). However even with an average succes rate of 20%, matching will still be higher than one (in years 3 to 5)														

To the Board of the Convergence of Health &
Technology Initiative

Direct dial +31 10 7032120
Fax number +31 10 7032302
Room number Na-1919
E-mail w.luiten@erasmusmc.nl
Our reference ML
Date March 11, 2022

Subject: Letter of commitment Leadership Flagship "Healthy Joints"

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

LS,

Street address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

With this letter I declare on behalf of the Department of General Practice that Prof. dr. Sita Bierma-Zeinstra, Head of Research at our department, will fully commit to the main leadership of the proposed flagship. She has a long and strong track record in leading, stimulating, merge, and accumulate multidisciplinary research in osteoarthritis.

I also fully support the Healthy Joints flagships' objective, which is to enable early stage personalized treatment or even preventive treatment of osteoarthritis. As osteoarthritis is the most common chronic and disabling disease in our society for which we only have limited effective treatment, the convergence in this flagship between health and technology is highly essential and needed to reach the long-term goal of adding at least five good quality-of-life years.

Head of Department
Prof.Dr. P.J.E. Bindels

Undergraduate Training
Dr. D. Pols

Institute for General Practice
Dr. J.M. van Ochten

Research
Mrs. Prof.Dr. S.M.A. Bierma-Zeinstra
Mrs. Dr. M. van Middelkoop

Institute for Intellectual
Disability Medicine
Mrs. Drs. J.M.T. Veeren

Intellectual Disability Medicine
Mrs. Dr. T.I.M. Hilgenkamp
Mrs. Dr. D.A.M. Maes-Festen

Sincerely yours,



Prof. dr. Patrick Bindels,
Head of the Department of General Practice